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DIRECTORATE-GENERAL FOR INFORMATICS

Directorate A – Digital Transformation
DIGIT.A.3 – Solutions for Legislation, Policy & HR



Getting started with LEOS

Technical guide

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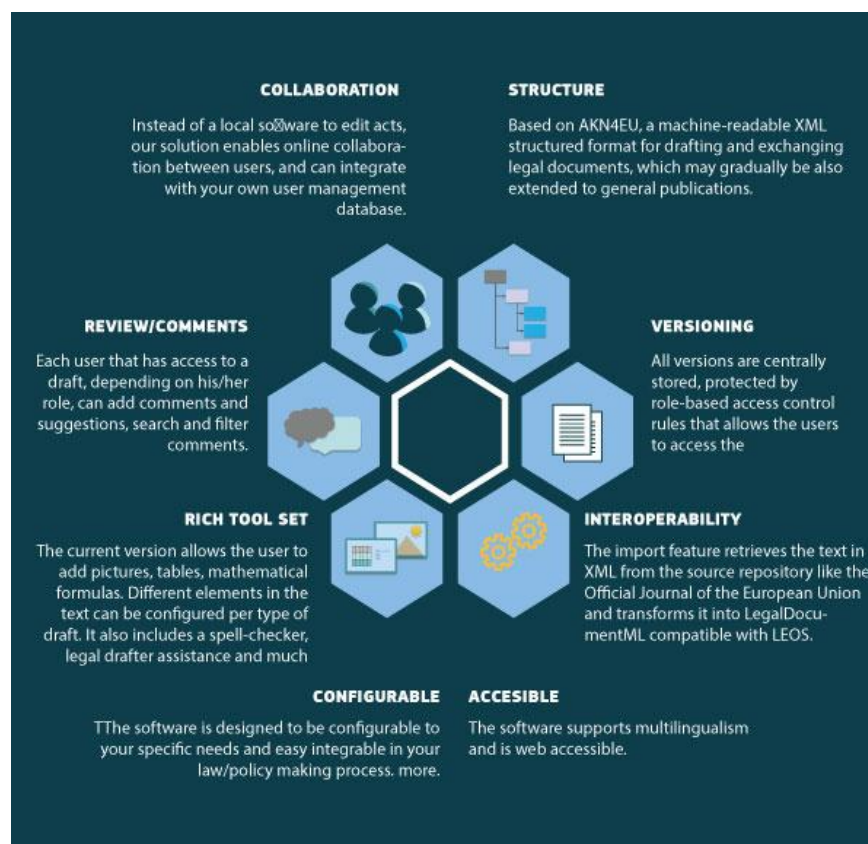
Introduction to LEOS

LEOS (Legislation Editing Open Software) is a project under [Interoperable Europe](#) initiative of the European Commission for a reinforced public sector interoperability policy, funded by the [Digital Europe Programme](#) (DIGITAL) and which was created to address the need of the public administration and European Institutions to generate draft legislation in a legal XML format.

The LEOS project under **Interoperable Europe** focuses on supporting the co-development co-design, and co-deployment of an 'IT eco-system centred on augmented LEOS'.

LEOS has been created to addresses the **modernisation and digital transformation of the drafting and revising of legislation in EU Institutions, EU agencies and bodies and Member States**.

LEOS ensures that the content drafted by the users follows the drafting guidelines by offering features like enforcing predefined document structures, predefined layout, numbering rules. All of this to ensure that the author would first and foremost focus on drafting itself and much less on layout management (or checking). To facilitate efficient online collaboration LEOS also contains some others features like comments, suggestions, version control, co-edition, etc.



Content is stored in an XML format, currently in AKN4EU which is based on the Akoma Ntoso OASIS standard namespace: <http://docs.oasis-open.org/legaldocml/ns/akn/3.0>. This facilitates document exchange, element retrieval, etc.

While Akoma Ntoso defines representations for parliamentary, legislative and judiciary documents, AKN4EU is a localisation of Akoma Ntoso and currently focus on the documents exchanged in the scope of the Ordinary Legislative Procedure (OLP).

- Regulation, Directive, Decision,
- Proposal for a Regulation, Proposal for a Directive, Proposal for a Decision.
- Some cases of the EP Texts Adopted.
- Council documents including the Position and the Statements of Council reasons.

(see AKN4EU 4.0 Baseline – General structure of documents, 0 Scope).

Each of the above legal documents have a certain structure, and it is usually composed of a cover page and one or more components. The components have their own structure.

Therefore, the main concepts LEOS is managing are the documents (the legal documents) and their components (or subdocuments).

Install LEOS

The current modules of augmented LEOS are:

- [LEOS core](#) and
- [Annotations utils](#).

IMPORTANT NOTES: *This release is intended to provide an experience with the software and is stripped of several important components to enable ease of use.*

Additionally, this software is adapted to run on a local server for demo purposes and without proper security mechanisms.

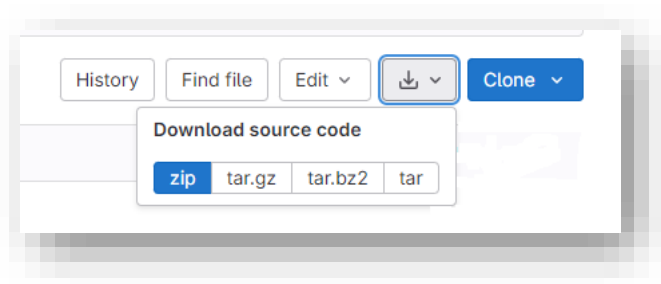
DEPENDENCIES

To compile the supplied source files and run the generated WAR the following software should be configured:

- Java SDK version 21.
- Maven version 3.8.X - 3.9.X (Maven runtime memory might need to be set: MAVEN_OPTS=-Xms256m -Xmx512m) (Maven settings, proxy and mirrors, might need to be adjusted to your environment and internet access requirements).
- Supported browser is Google Chrome version 45+ (Mozilla Firefox ESR version 38.3 and Microsoft Internet Explorer version 11 are known to work with minor issues).
- Node 20+.

1. Download the source code of both modules in the same folder.

Create a new directory {AUGMENTED_LEOS} in the local file system.

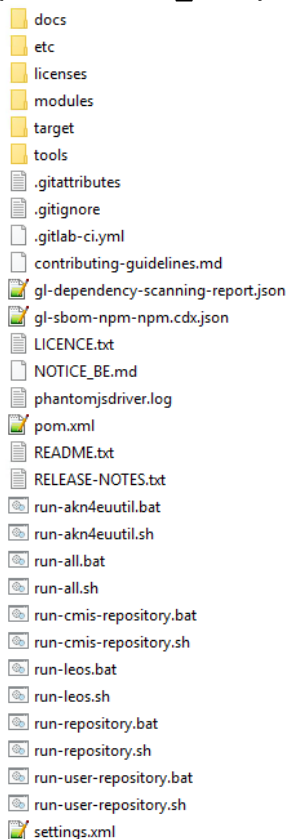


Download and unzip the following modules in the new directory {AUGMENTED_LEOS}:

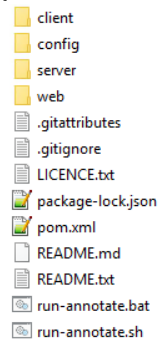
- [LEOS core](#)
- [Annotations utils](#)

Such that you'll end up with the following folder structure:

{AUGMENTED_LEOS}/leos



{AUGMENTED_LEOS}/annotate



2. Select demo installation option.

- If your machine's operating system is Microsoft Windows or Linux, you can use [Option A](#).
- If your machine's operating system is not windows or Linux based or you want to run components one by one, you should follow the steps of [Option B](#).

3. OPTION A

3.1. Run script.

Change the directory in the command line to be located /leos directory, in our example **{AUGMENTED_LEOS}/leos**.

If your machine's operating system is Microsoft Windows, you can simply execute the provided script:

```
run-all.bat.
```

Change the directory in the Command Prompt to be located /leos directory. In the example above it would be in { AUGMENTED_LEOS }/leos.

If your machine's operating system is Linux based, you can simply execute the provided script:

```
run-all.sh.
```

This script will execute individual scripts that will compile AND run each of the required software components.

If applications are not started using run-all.sh, open file and modify its content like below

```
./run-user-repository.sh &  
  
./run-repository.sh &  
  
./run-leos.sh &  
  
./run-akn4euutil.sh &
```

3.2. Use in browser.

Open the browser and navigate to the LEOS web interface available at the following URL:
<http://localhost:8080/leos-pilot/ui>

LEOS is pre-configured with these demo users:

NAME	LOGIN	PASSWORD	ROLE
Demo User	demo	demo	Normal
John Doe	john	demo	Normal
Jane Doe	jane	demo	Support
S Leo	leos	demo	Support

4. OPTION B

If your machine's operating system is not windows or Linux based or you want to run components one by one, you should follow below steps.

4.1. Running user database

You must compile and run the user database on the command line.

1. Traverse to folder {AUGMENTED_LEOS}/leos/tools/user-repo
2. Execute the following command to compile source code:

```
mvn clean install
```

3. Execute the following command to run it:

```
mvn spring-boot:run -Drun.profiles=h2
```

4.2. Running repository

- Use the InMemory repository version included with this LEOS distribution.

4.2.1. Use the InMemory repository version included with this LEOS distribution.

To run InMemory repository server, you must compile and run the repository on the command line.

- a. Traverse to folder {AUGMENTED_LEOS}/leos/tools/repository
- b. Execute the following command to clean source code.

```
mvn clean install
```

- c. Execute the following command to run it.

```
mvn spring-boot:run -Dspring-boot.run.additional-classpath-elements=../config/target/generated-config
```

- d. Normally, Access Inmemory repository console on web browser

<http://localhost:9097/repository/h2-console/>

4.2.2. Use external database with repository version included with this LEOS distribution.

- a. Configure which database is to be used in property file(s) in tools/repository/config/src/etc/config/filters

```

1 repository.db=h2
2
3 repository.jwt.auth.enabled=false
4 repository.jwt.auth.issuer.id=repository
5 repository.jwt.auth.access.token.expiration.min=600
6
7 repository.jwt.auth.user.details.cache.expiration.min=60
8 repository.jwt.auth.user.details.url=http://localhost:9095/ud-repo/users/{userId}
9
10 repository.jwt.auth.clients=leos
11 repository.jwt.auth.client.leos.id=leosClientId
12
13 repository.template.path=leos/templates
14 repository.template.custom.path=leos/templates/custom
15 repository.template.custom.version=1.0.0
16 repository.workspace.path=leos/workspaces/package_1
17 repository.data.script.path=data-property-h2.sql
18
19 repository.jwt.auth.issuer.secret=repositorySecret
20 repository.jwt.auth.client.leos.secret=leosSecret

```

if you want to add for instance oracle db, then just change repository.db=h2 --> repository.db=oracle

- b. Please make sure to add necessary dependencies in pom.xml for example if you want to add 'mysql' as repository db, then its necessary driver classes must be added.
- c. If you want to connect to oracle instance then, create a new property file named as 'repository-oracle.properties'

```

1 # Basic Spring Boot config for Oracle
2 spring.datasource.jndi-name=jdbc/leosRepository
3 spring.datasource.driver-class-name=oracle.jdbc.OracleDriver
4 spring.jpa.database-platform=org.hibernate.dialect.Oracle12cDialect
5
6 spring.datasource.url=jdbc:oracle:thin:@00.000.003.000:9999:oracle
7 spring.datasource.username=leos_repository
8 spring.datasource.password=leos_repository
9

```

- d. If you are running it locally then, JNDI name is not need, remove or comment property *spring.datasource.jndi-name*.
- e. If application is deployed on application server like Tomcat or Weblogic, then this JNDI name should match the JNDI name configured in application server.

4.3. Running LEOS

Notes:

- User database and repository must already be running.
- LEOS can use SAML protocol to authenticate with an IDP.
- To use SAML it is needed modify SAML configuration defined in "securityContext.xml" file inside module "security-saml".

You must run LEOS on the command line.

- 1) Traverse to folder {AUGMENTED_LEOS}/leos
- 2) Execute the following command to compile source code.

- a. For default authentication:

```
mvn clean install
```

- b. For SAML authentication:

```
mvn clean install Dsaml
```

- 3) Traverse to folder {AUGMENTED_LEOS}/leos/modules/web
- 4) Execute the following command to run LEOS.

- a. For default authentication:

```
mvn cargo:run | war
```

- b. For SAML authentication:

```
mvn cargo:run war Dsaml
```

4.4. RUNNING AKN4EUUTIL

You must compile and run akn4euutil on the command line.

- Traverse to folder { AUGMENTED_LEOS}/leos/tools/akn4euutil
- Execute the following command to compile source code.

```
mvn clean install
```

- Execute the following command to run it.

```
mvn spring boot:run -Dspring-boot.run.additional-classpath-elements=../config/target/generated-config
```

Open the browser and navigate to the LEOS web interface available at the following URL:

<http://localhost:8080/leos-pilot/ui>

4.5. VIEW H2 CONSOLE

1. Make sure *application.properties* has console enable:

```
spring.h2.console.enabled=true
```

2. Enter the below URL in browser (don't forget the context of the application)

- **USER REPO** : <http://localhost:9095/ud-repo/h2-console>
- **ANNOTATIONS**: <http://localhost:9099/annotate/h2-console>
- **JDBC URL** : jdbc:h2:mem:~/test;Mode=Oracle;DB_CLOSE_ON_EXIT=FALSE

5. Configure LEOS

More information on LEOS template and elements configuration can be found [here](#).